CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 58 (2017) 151–159

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic re-assessment of the Fe-Gd and Fe-Sm binary systems

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ARTICLE INFO

Keywords:
Fe-Gd and Fe-Sm binary systems
Phase diagram
Thermodynamics
CALPHAD

ABSTRACT

The Fe-Gd and Fe-Sm binary systems were re-assessed thermodynamically using the CALPHAD method based on the critical evaluation of previous optimizations and available experimental information in the published literature. The solution phases including liquid, fcc-Fe, bcc-Fe, bcc-Gd, hcp-Gd, bcc-Sm, hcp-Sm and rhombohedral-Sm, were described by the substitutional solution model and their excess Gibbs energies were expressed with the Redlich-Kister polynomial. The intermetallic compounds, α-$Fe_{17}Gd_2$, β-$Fe_{17}Gd_2$, $Fe_{23}Gd_6$, Fe_3Gd, Fe_2Gd, $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm, were modeled as stoichiometric compounds due to their narrow homogeneity ranges. Self-consistent thermodynamic parameters to describe Gibbs energies of various phases in the Fe-Gd and Fe-Sm binary systems were obtained finally. The calculated results in this work are in good agreement with the reported phase equilibria and thermodynamic properties.

1. Introduction

The rare-earth (RE) based permanent magnets, especially for Nd-Fe-B-based permanent magnets with excellent magnetic properties, have been used wildly in modern industry such as wind power, electric vehicles and hard disk [1–5]. In general, higher coercivity, higher maximum energy product and higher temperature stability of Nd-Fe-B-based permanent magnets were achieved greatly through the addition of heavy-rare-earth elements such as Dy and Tb [6–8]. Nd-Fe-B-based permanent magnets with a certain amount of Gd and/or Sm could affect their magnetic properties [9–11]. It indicates a potential way to improve the maximum operation temperature and magnetic properties of Nd-Fe-B magnets without heavy-rare-earth elements. In order to better understand the relationships between alloy composition, crystal structure, phase transformation and magnetic properties of Nd-Fe-B-based permanent magnets containing Gd and/or Sm elements, phase equilibria and thermodynamic properties of the Gd/Sm-Fe-B ternary systems are useful and indispensable.

As sub-binary systems in the Gd/Sm-Fe-B ternary systems, the Fe-Gd binary system was optimized by Liu et al. [12], Konar et al. [13], Zinkevich et al. [14–16] and Wang et al. [17]. Although the calculated temperatures of liquidus and invariant reactions in the Fe-Gd binary system by Liu et al. [12], Zinkevich et al. [14–16] and Wang et al. [17] are in agreement with experimental phase diagram data, the calculated enthalpies of mixing of liquid Fe-Gd alloys show obvious deviation from new experimental results measured by Ivanov et al. [18]. Using the modified quasichemical model, Konar et al. [13] optimized the Fe-Gd binary system considering new experimental data [18–20]. However, the polymorphic transformation, α-$Fe_{17}Gd_2$ (rhombohedral, $Zn_{17}Th_2$ type) ↔ β-$Fe_{17}Gd_2$ (hexagonal, $Ni_{17}Th_2$ type), at 1488 ± 2 K measured experimentally by Atiq et al. [21–23] was not taken into account in the previous calculations [12,13,17]. On the other hand, the Fe-Sm binary system was assessed thermodynamically by Zinkevich et al. [24]. The calculated Fe-Sm binary phase diagram is in good agreement with the experimental data [25]. However, the calculated enthalpies of mixing and partial enthalpies of mixing of Fe and Sm in liquid Fe-Sm alloys [24] are not consistent with the new experimental data measured by Berezutskii et al. [26]. Recently, the Fe-Sm binary system was re-optimized by Konar et al. [27,28] using the modified quasichemical model. Therefore, considering the compatibility of thermodynamic database of the RE-Fe-B multicomponent systems and new experimental results, thermodynamic re-assessment of the Fe-Gd and Fe-Sm binary systems are still necessary using the Thermo-calc® software.

In this work, based on the critical review of experimental phase diagram data and thermodynamic data in the reported literature, the Fe-Gd and Fe-Sm binary systems were re-assessed thermodynamically using the CALPHAD method.

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http://dx.doi.org/10.1016/j.calphad.2017.07.005
Received 19 May 2017; Received in revised form 12 July 2017; Accepted 14 July 2017
Available online 20 July 2017
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Table 1
Crystallographic data of stable phases in the Fe-Gd and Fe-Sm binary systems.

Phase	Prototype	Space group	Lattice parameters		References
			a (nm)	c (nm)	
δ-Fe	W	$Im\bar{3}m$	0.29315	—	[34]
γ-Fe	Cu	$Fm\bar{3}m$	0.36467	—	[34]
α-Fe	W	$Im\bar{3}m$	0.28665	—	[34]
β-Gd	W	$I\bar{3}m$	0.4060	—	[34]
α-Gd	Mg	$P63/mmc$	0.3634	0.7810	[34]
α-Sm	Sm	$R\bar{3}m$	0.3621	2.6255	[51]
β-Sm	Mg	$P63/mmc$	0.3642	0.5772	[51]
γ-Sm	La	$Im\bar{3}m$	0.3650	1.4156	[51]
α-Gd_2Fe_{17}	Th_2Zn_{17}	$R\bar{3}m$	0.8536	1.2429	[33]
β-Gd_2Fe_{17}	Th_2Ni_{17}	$P63/mmc$	0.8486	0.8349	[33]
Gd_6Fe_{23}	Th_6Mn_{23}	$Fm3m$	1.2134	—	[30]
$GdFe_3$	$NbBe_3$	$R\bar{3}m$	0.5166	2.4710	[30]
$GdFe_2$	$MgCu_2$	$Fd3m$	0.7390	—	[30]
Sm_2Fe_{17}	Th_2Zn_{17}	$R\bar{3}m$	0.8570	1.2430	[25]
$SmFe_3$	$NbBe_3$	$R\bar{3}m$	0.3190	2.4910	[25]
$SmFe_2$	$MgCu_2$	$Fd3m$	0.7417	—	[25]

2. Experimental information

2.1. The Fe-Gd binary system

The Fe-Gd binary phase diagram was investigated experimentally by Atiq et al. [21–23], Novy et al. [29], Savitskii et al. [30], Copeland et al. [31], Burov et al. [32] and Savistldi et al. [33]. Novy et al. [29] found seven intermetallic compounds, Fe_3Gd_2, Fe_2Gd, Fe_3Gd, Fe_7Gd_2, Fe_4Gd, Fe_5Gd and $Fe_{17}Gd_2$ in the Fe-Gd binary system and five invariant reactions were determined by differential thermal analysis (DTA) and optical microscopy (OM). Later, Savitskii et al. [30,33] investigated that four intermetallic compounds, $Fe_{17}Gd_2$, Fe_4Gd, Fe_3Gd and Fe_2Gd are formed through peritectic reactions measured by thermal analysis, OM and X-ray diffraction (XRD). Copeland et al. [31] and Burov et al. [32] indicated that three intermetallic compounds, Fe_3Gd_2, Fe_7Gd_2 and Fe_5Gd are formed through peritectic reactions using DTA, OM and XRD. Savitskii et al. [30,33], Copeland et al. [31] and Burov et al. [32] suggested that three intermetallic compounds, Fe_3Gd_2, Fe_7Gd_2 and Fe_5Gd, are not existent in the Fe-Gd binary system. In addition, Atiq et al. [21–23] measured the crystal structure transformation between α-$Fe_{17}Gd_2$ (rhombohedral, $Zn_{17}Th_2$ type) and β-$Fe_{17}Gd_2$ (hexagonal, $Ni_{17}Th_2$ type) at 1488 ± 2 K by DTA, dilatometry, and XRD. According to the experimental results [30–33], the Fe-Gd binary system was reviewed by Okamoto et al. [34,35] and Zhang et al. [36]. Okamoto et al. [34,35] and Zhang et al. [36] recommended that the intermetallic compound $Fe_{23}Gd_6$ is stable. Therefore, in the present work, five intermetallic compounds, α-$Fe_{17}Gd_2$, β-$Fe_{17}Gd_2$, $Fe_{23}Gd_6$, Fe_3Gd and Fe_2Gd, were taken into account. The crystal structure information of stable phases in the Fe-Gd binary system were summarized in Table 1.

The experimental results including the solidus, liquidus and invariant reactions determined by Savitskii et al. [30,33] and Atiq et al. [21–23] in the Fe-Gd binary system were accepted in this work. According to the experimental results [30,33], the temperature and composition of the eutectic reaction (L ↔ $GdFe_2$ + γ-Fe) are about 1103 K and 72 at% Gd, respectively. Four intermetallic compounds, $Fe_{17}Gd_2$, Fe_2Gd_{63}, Fe_3Gd and Fe_2Gd, are formed by peritectic reactions at 1608 K, 1553 K, 1433 K and 1353 K, respectively. Savitskii et al. [30] determined that the solubility of Gd in α-Fe is not more than 0.1 at% Gd at 1073 K, while the solubility of Fe in α-Gd is not more than 0.6 at % Fe at 973 K. Burov et al. [32] measured the solubility of Gd in α-Fe is not more than 0.1 at% Gd. The solubilities of Gd in α-Fe and γ-Fe were estimated by Savitskii et al. [30] to be 0.7 at% Fe and 0.5 at% Fe, respectively. In the present work, the solubilities of solid phases in the Fe-Gd binary system determined by Savitskii et al. [30] and Burov et al. [32] were considered.

The enthalpies of mixing of liquid Fe-Gd alloys at 1850 K were measured by Nikolaenko [37] by means of the isoperibolic calorimeter under high purity Ar gas. Recently, using high-temperature isoperibolic calorimeter under high purity He gas, Ivanov et al. [18] determined the enthalpies of mixing of liquid Fe-Gd alloys at 1833 K. The partial enthalpies of mixing of Fe and Gd in liquid Fe-Gd alloys were measured by Ivanov et al. [18] and Nikolaenko [37], respectively. The experimental results [18,37] show a large difference in terms of the location of minima of enthalpies of mixing and partial enthalpies of mixing of Fe and Gd. As can be shown from the experimental data [37], it is rather unusual to have minima of enthalpy of mixing where no solid phase exists. Furthermore, according to the enthalpy of mixing of the Fe-RE binary systems determined by Ivanov et al. [18], the systematic tendency of enthalpy of mixing of liquid Fe-RE alloys becomes much more negative, depending on the elements from La to Lu. The

Table 2
Thermodynamic parameters of the Fe-Gd binary system.

Phase	Thermodynamic parameters	Reference
Liquid	${}^{(0)}L_{Liq} = -18825.6670 + 9.5133T$	This work
	${}^{(1)}L_{Liq} = -3216.4805 + 5.7499T$	
bcc-A2	${}^{(0)}L^{bcc}_{Fe,Gd} = 32300$	This work
fcc-A1	${}^{(0)}L^{fcc}_{Fe,Gd} = 28000$	This work
hcp-A3	${}^{(0)}L^{hcp}_{Fe,Gd} = 30000$	This work
α-$Fe_{17}Gd_2$	$G_m^{\alpha-Fe_{17}Gd_2} = -210000 + 93.3217T + 17\,{}^0G^{bcc}_{Fe} + 2\,{}^0G^{hcp}_{Gd}$	This work
	$T_C^{Fe_{17}Gd_2} = 472\ \beta_0^{Fe_{17}Gd_2} = 21.2$	[52]
β-$Fe_{17}Gd_2$	$G_m^{\beta-Fe_{17}Gd_2} = -210000 + 93.3181T + 17\,{}^0G^{bcc}_{Fe} + 2\,{}^0G^{hcp}_{Gd}$	This work
$Fe_{23}Gd_6$	$G_m^{Fe_{23}Gd_6} = -395000 + 154.5061T + 23\,{}^0G^{bcc}_{Fe} + 6\,{}^0G^{hcp}_{Gd}$	This work
	$T_C^{Fe_{23}Gd_6} = 468\ \beta_0^{Fe_{23}Gd_6} = 14.8$	[52]
Fe_3Gd	$G_m^{Fe_3Gd} = -54800 + 20.8834T + 3\,{}^0G^{bcc}_{Fe} + {}^0G^{hcp}_{Gd}$	This work
	$T_C^{Fe_3Gd} = 728\ \beta_0^{Fe_3Gd} = 1.6$	[52]
Fe_2Gd	$G_m^{Fe_2Gd} = \begin{cases} -37700 - 5T - 1.825\times10^{-3}T^3 + 4.65\times10^{-6}T^4 & (T < 120) \\ \left.\begin{matrix} -39500 + 91.6827T - 12T\ln T + 0.011T^2 \\ -365000T^{-1} - 3\times10^{-6}T^3 + 2\,{}^0G^{bcc}_{Fe} + {}^0G^{hcp}_{Gd} \end{matrix}\right\} & (T > 120) \end{cases}$	This work
	$T_C^{Fe_2Gd} = 782\ \beta_0^{Fe_2Gd} = 3.35$	[52]

Table 3
Thermodynamic parameters of Fe-Sm binary system.

Phase	Thermodynamic parameters	Reference
Liquid	${}^{(0)}L_{Liq} = -16522.9530 + 16.8415T$	This work
	${}^{(1)}L_{Liq} = -4350.2890 + 7.7475T$	
bcc-A2	${}^{(0)}L^{bcc}_{Fe,Sm} = 100000$	This work
hcp-A3	${}^{(0)}L^{hcp}_{Fe,Sm} = 100000$	This work
fcc-A1	${}^{0}G^{fcc}_{Fe}$ cited from SGTE database	[50]
rhombohedral	${}^{0}G^{rohmbohedral}_{Sm}$ cited from SGTE database	[50]
$Fe_{17}Sm_2$	$G^{Fe_{17}Sm_2}_m = \begin{cases} -153289 + 85T - 8.25\times10^{-3}T^3 + 1.74\times10^{-5}T^4 & (T<120) \\ \left.\begin{matrix}-210000 + 2585.0031T - 452.5T\ln T - 0.19T^2 \\ +1050000T^{-1} + 5.5\times10^{-6}T^3\end{matrix}\right\} & (T>120)\end{cases}$	This work
	$T^{Fe_{17}Sm_2}_C = 394\ \beta^{Fe_{17}Sm_2}_0 = 34.2$	[24]
Fe_3Sm	$G^{Fe_3Sm}_m = -38800 + 21.3482T + 3\,{}^{0}G^{bcc}_{Fe} + {}^{0}G^{rohmbohedral}_{Sm}$	This work
	$T^{Fe_3Sm}_C = 653\ \beta^{Fe_3Sm}_0 = 4.6$	[53]
Fe_2Sm	$G^{Fe_2Sm}_m = -25500 + 12.8266T + 2\,{}^{0}G^{bcc}_{Fe} + {}^{0}G^{rohmbohedral}_{Sm}$	This work
	$T^{Fe_2Sm}_C = 682\ \beta^{Fe_2Sm}_0 = 2.68$	[54]

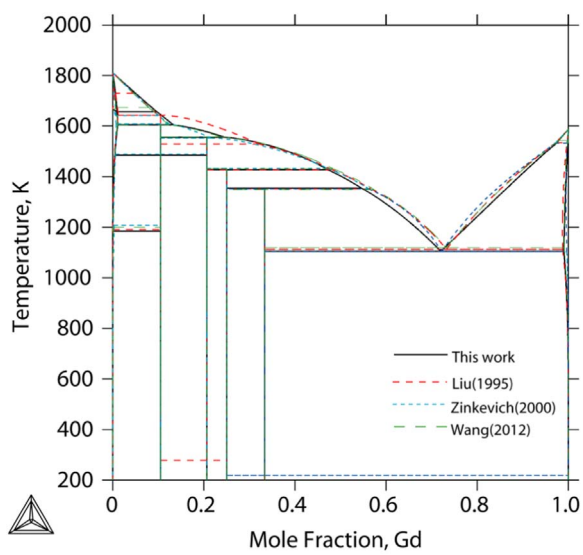

Fig. 1. Calculated phase diagram of the Fe-Gd binary system in this work with the calculated results [12,14,17].

enthalpies of mixing of liquid Fe-Gd alloys determined by Ivanov et al. [18] at 1833 K and Nikolaenko [37] at 1850 K should be less positive than those of liquid Fe-Tb alloys at 1833 K and liquid Fe-Dy alloys at 1850 K determined by Ivanov et al. [18]. However, the enthalpies of mixing of liquid Fe-Gd alloys at 1850 K determined by Nikolaenko [37] are much more negative than those of liquid Fe-Dy alloys at 1850 K determined by Ivanov et al. [18]. In addition, the experimental error of partial enthalpies of mixing determined by Nikolaenko [37] is obviously larger than that measured by Ivanov et al. [18]. Therefore, the experimental results [18] over the whole composition range are more reliable and were used in the present optimization.

Colinet and Pasturel [38] determined the enthalpies of formation of intermetallic compounds α-$Fe_{17}Gd_2$, Fe_3Gd and Fe_2Gd at 298 K to be −2.3 ± 1.6, −9.3 ± 1.9 and −11.6 ± 2 kJ/mol-atom using aluminum solution calorimetry. Deodhar and Ficalora [39] measured the enthalpy of formation of Fe_2Gd to be −3.5 ± 1.5 kJ/mol-atom at 298 K by differential thermal analysis. The enthalpy of formation of Fe_2Gd measured by Deodhar and Ficalora [39] is less negative than the experimental data [38]. Recently, Meschel et al. [20] measured the enthalpy of formation of Fe_2Gd to be −1.8 ± 3.4 kJ/mol-atom by directed synthesis calorimetry. In their experiment [20], the Fe_2Gd sample contains up to 5 wt% $Fe_{17}Gd_2$. It is believed that the experimental values determined by Deodhar and Ficalora [39] and Meschel et al. [20] are less reliable. Using the electromotive force of solid-state galvanic cells under vapor pressure, the enthalpy of formation of $Fe_{17}Gd_2$ measured by Gozzi et al. [19] at 973 K is −3 ± 0.03 kJ/mol-atom, which is similar with the experimental data [38]. In addition, Germano and Butera [40] determined the heat capacity, entropy of formation (S^o_T) and the enthalpy difference referred to 0 K ($H^o_T - H^o_0$) of Fe_2Gd by adiabatic calorimetry in the low temperature range (8–300 K). The experimental data [19,38,40] are much more reliable and were taken into account in the present optimization.

2.2. The Fe-Sm binary system

The Fe-Sm binary phase diagram was investigated experimentally by Buschow et al. [25] in the whole composition rang using XRD, DTA and metallographic. Three intermetallic compounds, $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm were found. Three peritectic reactions, L + γ-Fe ↔ $Fe_{17}Sm_2$ at 1559 K, L + $Fe_{17}Sm_2$ ↔ Fe_3Sm at 1287 K, L + Fe_3Sm ↔ Fe_2Sm at 1173 K and the eutectic reaction, L ↔ Fe_2Sm + α-Sm at 997 K were determined. Based on the experimental results [25], the Fe-Sm binary system was reviewed by Okamoto [41]. The crystal structures of $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm in the Fe-Sm binary system were reported by Johnson et al. [42], Nassau et al. [43], Wernick et al. [44] and Mansey et al. [45]. According to phase relations of the Fe-Sm-X (X = B, Ti. Bi, Sn) ternary systems investigated by Liu et al. [46], Stepien-Damm et al. [47], Bodak et al. [48] and Xu et al. [49], respectively, Fe_2Sm, Fe_3Sm and $Fe_{17}Sm_2$ were also confirmed to be stable. The crystal structure information of the stable phases in the Fe-Sm binary system was shown in Table 1. In addition, the solubilities of Fe and Sm in terminal solid phases (bcc-Fe, fcc-Fe, hcp-Sm, bcc-Sm and rhombohedral-Sm) were not taken into account in this work due to the lack of available experimental data.

The enthalpies of mixing and partial enthalpy of mixing of Sm and Fe in liquid Fe-Sm alloys at 1829 K in the entire composition range was measured by Berezutskii et al. [26] using high temperature calorimetric method. Meschel et al. [20] measured the enthalpy of formation of intermetallic compound $Fe_{17}Sm_2$ to be −3.1 ± 3.2 kJ/mol-atom by directed synthesis calorimetry. Using the physical property measurement system, Zinkevich et al. [24] determined the heat capacity of $Fe_{17}Sm_2$ in the low temperature range (5–298 K) and its entropy of formation to be 36.6 ± 1.5 J/mol-K-atom at 298 K. The experimental results [20,24,26] mentioned above were used in the present optimization.

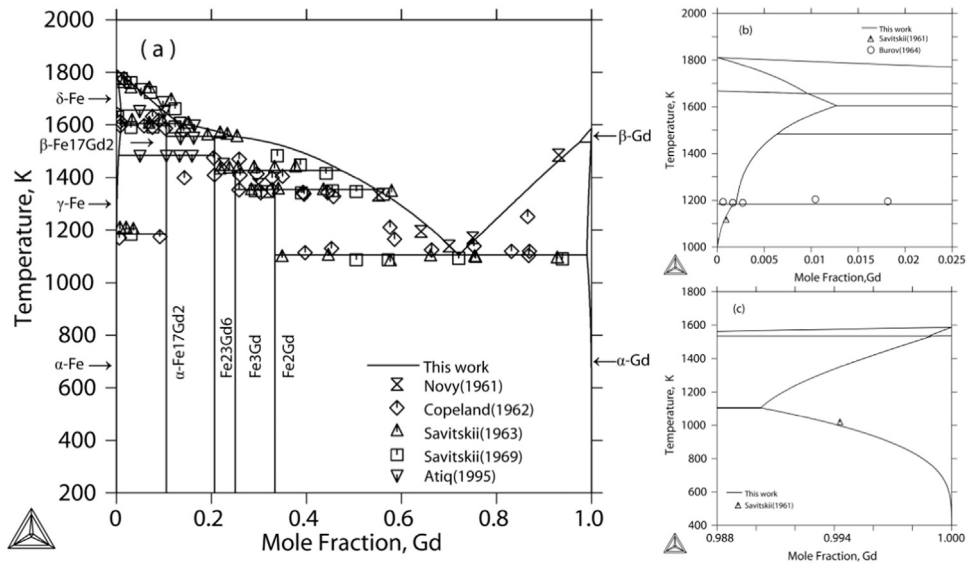

Fig. 2. Calculated Fe-Gd phase diagram of the Fe-Gd system with the experimental data [23,29–33].

3. Thermodynamic modeling

3.1. Pure elements

The Gibbs energy of pure elements in the Fe-Gd and Fe-Sm binary systems can be given as:

$$^{0}G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

in which φ stands for the status of the element i and T is the absolute temperature in Kelvin. $G_i^{\varphi}(T)$ is the Gibbs energy of the element i with the structure ϕ.H_i^{SER} is the enthalpy of the element i in the standard reference state (298.15 K, 1 bar). $^{0}G_i^{\phi}(T)$ is the molar Gibbs energy of the element i with the structure ϕ referred to the enthalpy of its stable state at 298.15 K and 1 bar. In the present work, $^{0}G_{Fe}^{\varphi}(T)$, $^{0}G_{Gd}^{\varphi}(T)$ and $^{0}G_{Sm}^{\varphi}(T)$ are taken from the Scientific Group Thermodata Europe (SGTE) database compiled by Dinsdale [50].

3.2. Solid solution phases

The solution phase ϕ(ϕ= liquid, bcc-Fe, fcc-Fe, hcp-Gd, bcc-Gd, hcp-Sm, bcc-Sm and rhombohedral-Sm) in the Fe-Gd and Fe-Sm binary systems is described with the substitutional solution model, and the Gibbs energies of these solution phases are expressed as follows:

$$G_m^{\varphi} = \sum_{i=Fe,RE} {}^{0}G_i^{\varphi}x_i + RT\sum_{i=Fe,RE} x_i\ln x_i + x_{Fe}x_{RE}\sum_{j=0}^{n} {}^{j}L_{Fe,RE}^{\varphi}(x_{Fe} - x_{RE})^j + {}^{mag}G_m^{\varphi} \quad (2)$$

$$^{mag}G_m^{\varphi} = RT\ln(\beta_0 + 1)g(\tau) \quad (3)$$

where $^{0}G_i^{\phi}$ is the molar Gibbs free energy of the element i in the respective reference state with the structure ϕ in the nonmagnetic state and is taken from the database compiled by Dinsdale [50]. x_i is the mole fraction of the element i, R is the gas constant and T is the absolute temperature. $^{mag}G_m^{\varphi}$ is the magnetic contribution to the Gibbs energy of the phase ϕ. τ is the normalized temperature as expressed as: $\tau = T/T_C^{\varphi}$, and T_C^{φ} is the Curie temperature of the phase φ. β_0 is the Bohr magnetons (per mole atoms). $^{j}L_{Fe,RE}^{\varphi}$ is expressed as:

$$^{j}L_{Fe,RE}^{\varphi} = A_j + B_jT \quad (4)$$

where A_j and B_j are parameters to be optimized in the present work. In addition, according the equation suggested by Hillert and Jarl [51], $g(\tau)$ is expressed as:

$$g(\tau) = 1 - \frac{1}{M}\left[\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p} - 1\right)\left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600}\right)\right] \quad (\tau \le 1) \quad (5)$$

$$g(\tau) = -\frac{1}{M}\left(\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500}\right) \quad (\tau > 1) \quad (6)$$

$$M = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p} - 1\right) \quad (7)$$

where the parameter p is determined by the structure of the phase ϕ (p = 0.4 for bcc structure and 0.28 for other structures).

Note that the lattice stability of fcc-Gd is not available in the SGTE database given by Dinsdale [50]. In the present work, it is taken from the data reported by Liu et al. [12] as:

$$^{0}G_{Gd}^{fcc} = 5000 + {}^{0}G_{Gd}^{hcp} \quad (8)$$

3.3. Intermetallic phases

The intermetallic compounds, α-$Fe_{17}Gd_2$, β-$Fe_{17}Gd_2$, $Fe_{23}Gd_6$, Fe_3Gd, Fe_2Gd, $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm in the Fe-Gd and Fe-Sm binary systems are treated as stoichiometric compound. Because the heat capacities of Fe_2Gd and $Fe_{17}Sm_2$ were determined experimentally by Germano et al. [40] and Zinkevich et al. [24], respectively, their Gibbs energies are expressed as:

$$G_m^{Fe_2Gd} = \begin{cases} C_1 + D_1T + E_1T^3 + F_1T^4 + {}^{mag}G_m^{Fe_2Gd} & (T < 120) \\ 2\,{}^{0}G_{Fe}^{bcc} + {}^{0}G_{Gd}^{hcp} + C_2 + D_2T + E_2T\ln T \\ +F_2T^2 + H_1T^{-1} + I_1T^3 + {}^{mag}G_m^{Fe_2Gd} \end{cases} (T > 120) \quad (9)$$

$$G_m^{Fe_{17}Sm_2} = \begin{cases} C_3 + D_3T + E_3T^3 + F_3T^4 + {}^{mag}G_m^{Fe_{17}Sm_2} & (T < 120) \\ 17\,{}^{0}G_{Fe}^{bcc} + 2\,{}^{0}G_{Sm}^{rohmbohedral} + C_4 + D_4T + E_4T\ln T \\ +F_4T^2 + H_2T^{-1} + I_2T^3 + {}^{mag}G_m^{Fe_{17}Sm_2} \end{cases} (T > 120) \quad (10)$$

The parameters C_i, D_i, E_i, F_i, H_i and I_i, (i= 1,2,3,4) are to be optimized in this work. Due to the lack of heat capacity, their Gibbs

Table 4
Invariant reactions in the Fe-Gd binary system.

Reaction	Type	T/K	Composition (x_{Gd}^{L})	Reference
L + δ-Fe ↔ γ-Fe	Peritectic	1653.0	0.090	[30]
		1661 ± 5	—	[31]
		1653.0	0.090	[12]
		1654.0	0.147	[14]
		1652.0	0.090	[14]
		1673.0	0.080	[17]
		1657.0	0.090	This work
L + γ-Fe ↔ β-$Fe_{17}Gd_2$	Peritectic	1604.0	0.119	[30]
		1593 ± 5	—	[31]
		1604.0	0.120	[12]
		1605.0	0.191	[13]
		1607.0	0.124	[14]
		1604.4	0.123	[17]
		1604.0	0.124	This work
L + β-$Fe_{17}Gd_2$ ↔ $Fe_{23}Gd_6$	Peritectic	1556.0	0.234	[30]
		1553 ± 5	—	[31]
		1556.0	0.230	[12]
		1550.0	0.281	[13]
		1552.9	0.215	[14]
		1557.1	0.236	[17]
		1556.0	0.269	This work
L + $Fe_{23}Gd_6$ ↔ Fe_3Gd	Peritectic	1429.0	0.477	[30]
		1423 ± 5	—	[31]
		1429.0	0.470	[12]
		1439.0	0.432	[13]
		1431.0	0.474	[14]
		1431.5	0.483	[17]
		1429.0	0.490	This work
L + Fe_3Gd ↔ Fe_2Gd	Peritectic	1355.0	0.560	[30]
		1353.0	—	[31]
		1355.0	0.560	[12]
		1347.0	0.510	[13]
		1350.5	0.564	[14]
		1348.0	0.573	[17]
		1354.5	0.565	This work
L ↔ Fe_2Gd + α-Gd	Eutectic	1105.0	0.736	[30]
		1118 ± 5	—	[31]
		1105.0	0.735	[12]
		1113.0	0.673	[13]
		1101.9	0.713	[14]
		1119.0	0.731	[17]
		1105.0	0.736	This work
β-$Fe_{17}Gd_2$ ↔ α-$Fe_{17}Gd_2$		1488 ± 2	—	[23]
		1488.4	—	[14]
		1486.0	—	This work
L + β-Gd ↔ α-Gd	Peritectic	1512.0	0.960	[30]
		1508.0	—	[31]
		1512.0	0.960	[12]
		1428.0	0.896	[13]
		1532.0	0.970	[14]
		1543.2	0.980	[17]
		1512.0	0.970	This work

energies of α-$Fe_{17}Gd_2$, β-$Fe_{17}Gd_2$, $Fe_{23}Gd_6$, Fe_3Gd, Fe_3Sm and Fe_2Sm are expressed using the Neumann-Kopp rule as:

$$G_m^{\alpha-Fe_{17}Gd_2} = 17\,{}^0G_{Fe}^{bcc} + 2\,{}^0G_{Gd}^{dhcp} + J_1 + K_1T + {}^{mag}G_m^{\alpha-Fe_{17}Gd_2} \quad (11)$$

$$G_m^{\beta-Fe_{17}Gd_2} = 17\,{}^0G_{Fe}^{bcc} + 2\,{}^0G_{Gd}^{dhcp} + J_2 + K_2T + {}^{mag}G_m^{\beta-Fe_{17}Gd_2} \quad (12)$$

$$G_m^{Fe_{23}Gd_6} = 23\,{}^0G_{Fe}^{bcc} + 6\,{}^0G_{Gd}^{dhcp} + J_3 + K_3T + {}^{mag}G_m^{Fe_{23}Gd_6} \quad (13)$$

$$G_m^{Fe_3Gd} = 3\,{}^0G_{Fe}^{bcc} + {}^0G_{Gd}^{dhcp} + J_4 + K_4T + {}^{mag}G_m^{Fe_3Gd} \quad (14)$$

$$G_m^{Fe_3Sm} = 3\,{}^0G_{Fe}^{bcc} + {}^0G_{Sm}^{rohmbohedral} + J_5 + K_5T + {}^{mag}G_m^{Fe_3Sm} \quad (15)$$

$$G_m^{Fe_2Sm} = 2\,{}^0G_{Fe}^{bcc} + {}^0G_{Sm}^{rohmbohedral} + J_6 + K_6T + {}^{mag}G_m^{Fe_2Sm} \quad (16)$$

The parameters J_i and K_i (= 1, 2, 3, 4, 5, 6) are to be optimized in this work. ${}^{mag}G_m^{\varphi}$ (φ = α-$Fe_{17}Gd_2$, β-$Fe_{17}Gd_2$, $Fe_{23}Gd_6$, Fe_3Gd, Fe_2Gd, $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm) is expressed using the Eq. (3). In the present work, Curie temperature (T_C^{φ}) and Bohr magneton (β_0) of all

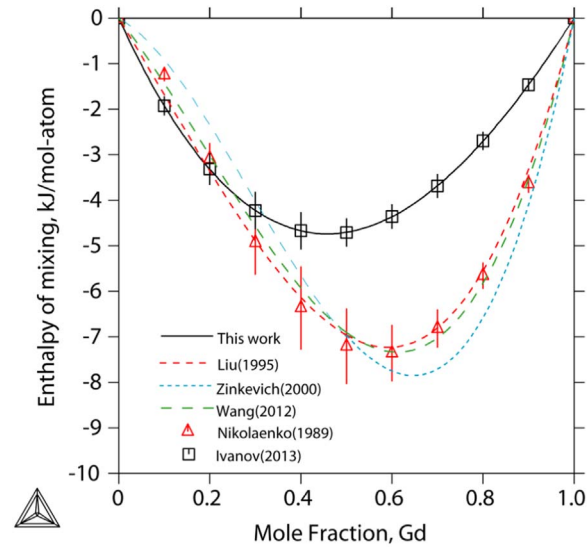

Fig. 3. Calculated enthalpy of mixing of liquid Fe-Gd alloys at 1833 K with the experimental data [18,37] and the calculated results [12,14,17]. (Reference states: liquid Fe and liquid Gd).

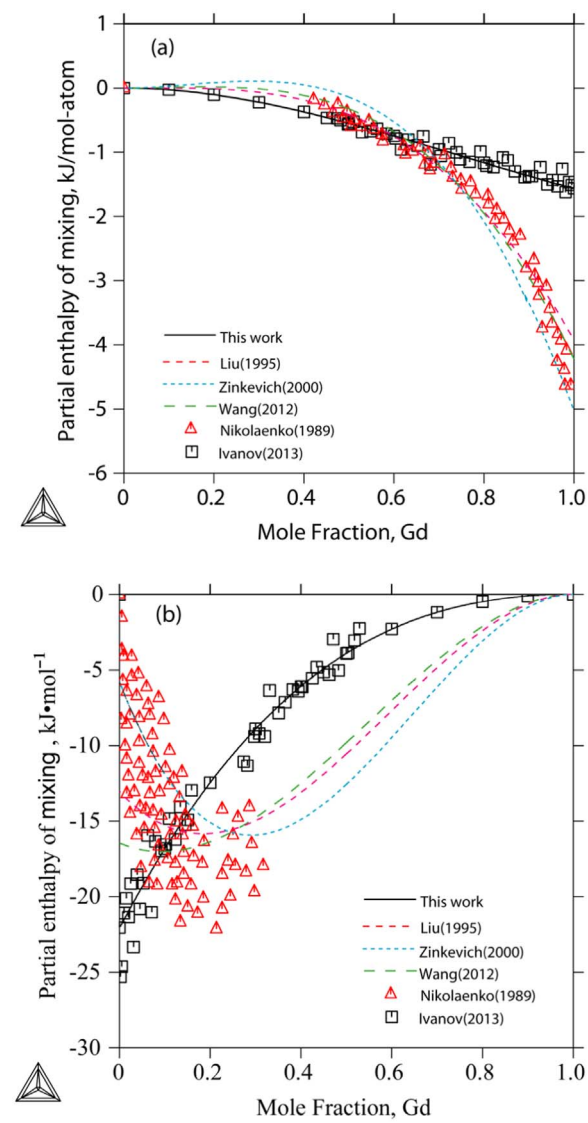

Fig. 4. Calculated partial enthalpy of Fe and Gd in liquid Fe-Gd alloys at 1833 K with the experimental data [18,37] and calculated results [12,14,17]. (Reference states: liquid Fe and liquid Gd).

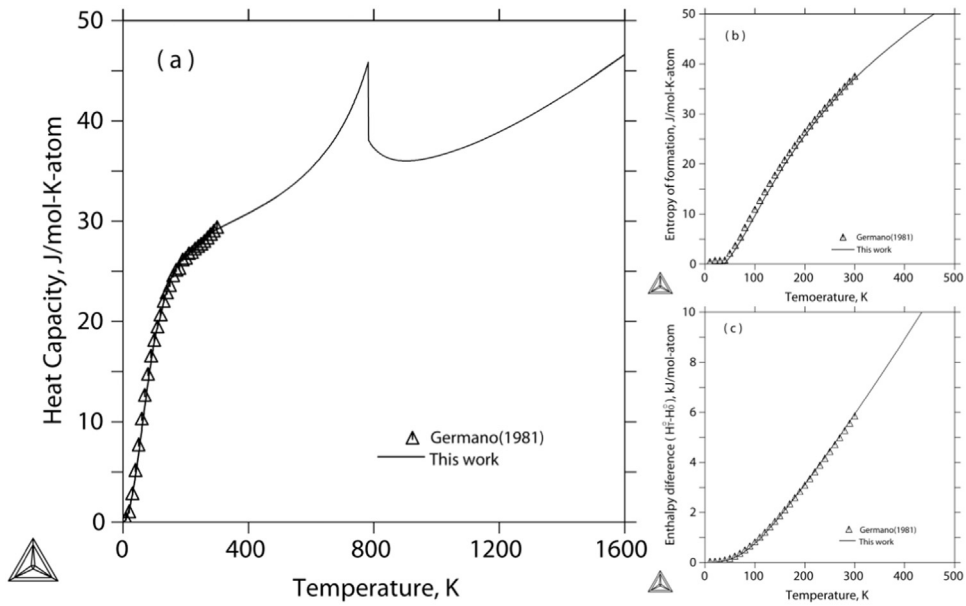

Fig. 5. Calculated heat capacity, entropy of formation (S_f^o) and enthalpy difference ($H_f^o - H_0^o$) of Fe_2Gd with the experimental data [40].

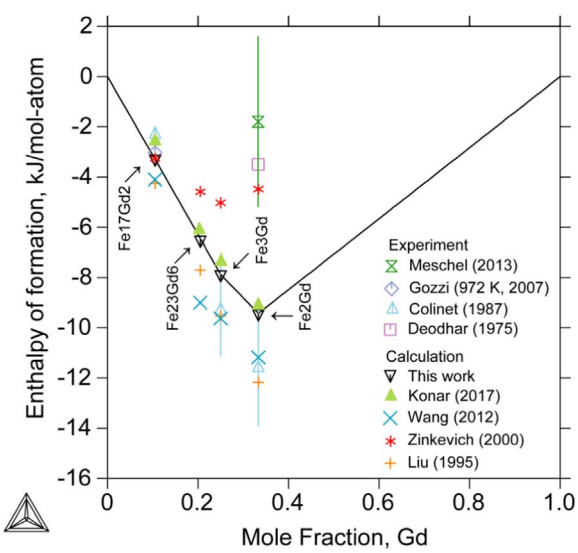

Fig. 6. Calculated enthalpies of formation of intermetallic compounds at 298 K with the experimental data [19,20,38,39] and the calculated results [12–14,17]. (Reference states: α-Fe and α-Gd).

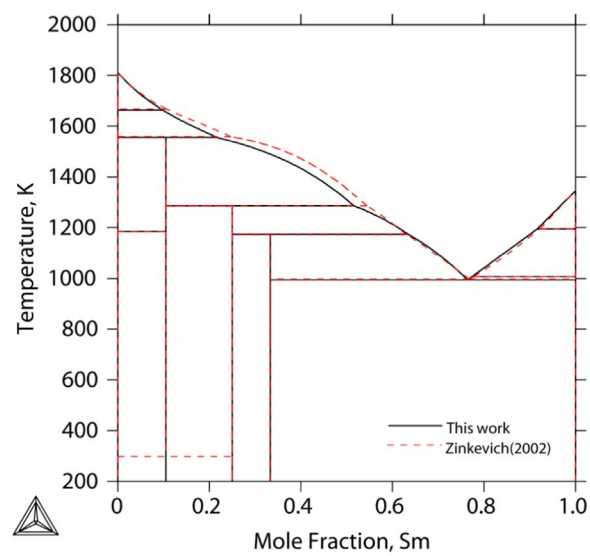

Fig. 7. Calculated phase diagram of the Fe-Sm binary system in this work with the calculated results [24].

intermetallic compounds in the Fe-Gd and Fe-Sm binary systems were used as shown in Table 2 and Table 3, respectively.

4. Results and discussion

Based on the experimental data reported in the literature and the stabilities of the elements Fe, Gd and Sm compiled by Dinsdale [50], thermodynamic parameters of various phases in the Fe-Gd and Fe-Sm binary systems were optimized using the PARROT module in the Thermo-calc® software package developed by Sundman et al. [55]. The PARROT module works by minimizing the square sum of the differences between the experimentally determined data and the calculated values. During the optimizing procedure, each set of the experimental data is given the certain weight according to its reliability and compatibility of the experimental data. It should be mentioned that the more reliable the experimental data are the higher weight will be set. The weight of each experimental data can be changed systematically during the optimization until most of the experimental data can be taken into consideration within the claimed uncertainty limits. Finally, thermodynamic parameters of all phases obtained in the Fe-Gd and Fe-Sm binary systems were given in Table 2 and Table 3, respectively. The calculated results with the experimental data were discussed in detail in the following section.

4.1. The Fe-Gd binary system

Fig. 1 and Fig. 2 are the calculated Fe-Gd binary phase diagram in this work with the calculated results [12,14,17] and the experimental data [23,29–33], respectively. As can be seen in Fig. 1, the intermetallic compounds, α-$Fe_{17}Gd_2$, $Fe_{23}Gd_6$, Fe_3Gd and Fe_2Gd are stable at room temperature in the present work. The calculated liquidus are in good agreement with the experimental data determined by Savitskii et al. [30,33], Novy et al. [29] and Atiq et al. [23] as shown in Fig. 2(a). Fig. 2(b) and Fig. 2(c) compare the calculated phase boundary in the Fe-rich and Gd-rich side with the experimental results reported by Savitskii et al. [30] and Burov et al. [32], respectively. The calculated solubility of Gd in bcc-Fe (less 0.1 at% Gd at 1073 K) and that of Fe in

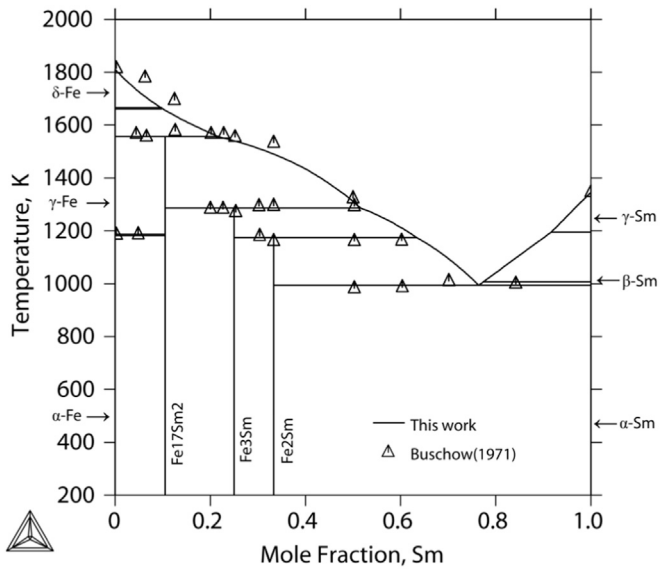

Fig. 8. Calculated phase diagram of the Fe-Sm binary system with the experimental data [25].

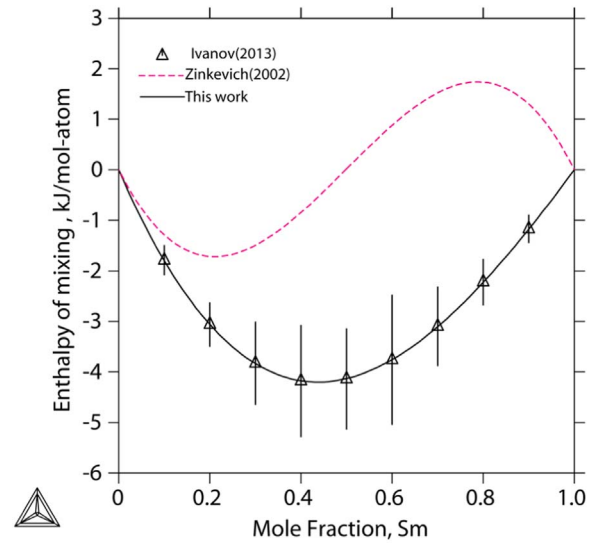

Fig. 9. Calculated enthalpy of mixing of liquid Fe-Sm alloys at 1829 K with the experimental data [26] and calculated results [24]. (Reference states: liquid Fe and liquid Sm).

Table 5
Invariant reactions in the Fe-Sm binary system.

Reaction	Type	T/K	Composition (x^L_{Sm})	Reference
L + δ-Fe ↔ γ-Fe	Peritectic	1667.0	0.110	[25]
		1667.5	0.106	[24]
		1664.0	0.110	This work
L + γ-Fe ↔ $Fe_{17}Sm_2$	Peritectic	1559.0	0.250	[25]
		1558.5	0.247	[24]
		1557.0	0.238	This work
L + $Fe_{17}Sm_2$ ↔ Fe_3Sm	Peritectic	1287.0	0.510	[25]
		1287.0	0.544	[24]
		1287.0	0.527	This work
L + Fe_3Sm ↔ Fe_2Sm	Peritectic	1173.0	0.610	[25]
		1172.5	0.631	[24]
		1174.0	0.611	This work
L ↔ Fe_2Sm + α-Sm	Eutectic	997.0	0.700	[25]
		997.0	0.768	[24]
		995.8	0.730	This work
L + γ-Sm ↔ β-Sm	Peritectic	1195.0	0.920	[25]
		1195.0	0.922	[24]
		1195.0	0.906	This work
L + β-Sm ↔ α-Sm	Peritectic	1007.0	0.780	[25]
		1007.0	0.777	[24]
		1007.0	0.773	This work

hcp-Gd (less 0.6 at% Fe at 973 K) were reproduced well with the experimental data [30]. The calculated temperatures and compositions of invariant reactions in the Fe-Gd binary system with the experimental data were summarized in Table 4. In this work, the temperature of the eutectic reaction, L ↔ Fe_2Gd+β-Gd, was calculated to be 1105.0 K, while the calculated temperatures of four peritectic reactions, L + γ-Fe ↔ $Fe_{17}Gd_2$, L + $Fe_{17}Gd_2$ ↔ $Fe_{23}Gd_6$, L + $Fe_{23}Gd_6$ ↔ Fe_3Gd and L + Fe_3Gd ↔ Fe_2Gd, are 1604.0 K, 1556.0 K, 1428.3 K and 1354.8 K, which are in good agreement with the experimental results [23,29–33]. The polymorphic transformation, α-$Fe_{17}Gd_2$ ↔ β-$Fe_{17}Gd_2$, was calculated to be 1486 K in this work, which agrees with the data (1488 ± 2 K) measured by Atiq et al. [21–23]. Furthermore, as can be seen in Table 4, the calculated compositions of invariant reactions in the Fe-Gd binary system in the present work are consistent with the experimental results measured by Savitskii et al. [30,33] and Atiq et al. [23].

Fig. 3 is the calculated enthalpy of mixing of liquid Fe-Gd alloys at 1833 K with the experimental data determined by Ivanov et al. [18] and the calculated results [12,14,17]. The partial enthalpies of mixing of Fe and Gd in liquid Fe-Gd alloys at 1833 K compared with the experimental data measured by Ivanov et al. [18] and the calculated results [12,14,17] as show in Fig. 4. The enthalpy of mixing and the partial enthalpy of mixing of liquid Fe-Gd alloys assessed previously by Liu et al. [12], Zinkevich et al. [14] and Wang et al. [17] using the experimental data [37] shows much more negative than the latest experimental data [24]. In the present work, the calculated results are in good agreement with the latest experimental data [24].

Fig. 5 is the calculated the heat capacity, the entropy of formation (S^o_T) and the enthalpy difference referred to 0 K ($H^o_T - H^o_0$) of the intermetallic compound Fe_2Gd with the experimental data [40]. The calculated heat capacity of Fe_2Gd is consistent with the experimental data measured by Germano and Butera [40] in Fig. 5(a). The calculated entropy of formation of Fe_2Gd at 298 K (S^o_{298}) is 36.890 J/mol-K-atom, which is in agreement with the experimental data (38.627 J/mol-K-atom) [40]. The calculated enthalpy difference ($H^o_{298} - H^o_0$) of Fe_2Gd is −5.875 kJ/mol-atom, which is in good accordance with the experimental value (−5.754 kJ/mol-atom) [40]. In Fig. 5(b) and Fig. 5(c), the calculated entropy of formation (S^o_T) and the enthalpy difference referred to 0 K ($H^o_T - H^o_0$) of Fe_2Gd agree well with the experimental data [40]. The Gibbs energy difference ($G^o_{298} - H^o_0$) of Fe_2Gd was calculated to be −5.117 kJ/mol-atom, which is slightly different with the experimental value (−5.757 kJ/mol-atom) [40].

Fig. 6 is the calculated enthalpies of formation of intermetallic compounds in the Fe-Gd binary system at 298 K with the experimental data [19,20,38,39]. The calculated enthalpy of formation of $Fe_{17}Gd_2$ is in good agreement with the experimental data determined by Colinet and Pasturel [38] and is slightly more negative than the data measured by Gozzi et al. [19]. The calculated enthalpy of formation of Fe_3Gd agrees well with the experimental data [38], while the enthalpy of formation of Fe_2Gd is less negative than the experimental data [38]. According to the systemic analysis [13] in the assessment of the Fe-RE binary systems, the enthalpies of formation of intermetallic compounds should be more negative, depending on the heavy rare earth element from Gd to Lu. Therefore, the calculated enthalpies of formation of intermetallic compounds in the Fe-Gd binary system are reasonable.

4.2. The Fe-Sm binary system

Fig. 7 is the calculated phase diagram of the Fe-Sm binary system in this work with the calculated results reported by Zinkevich et al. [24].

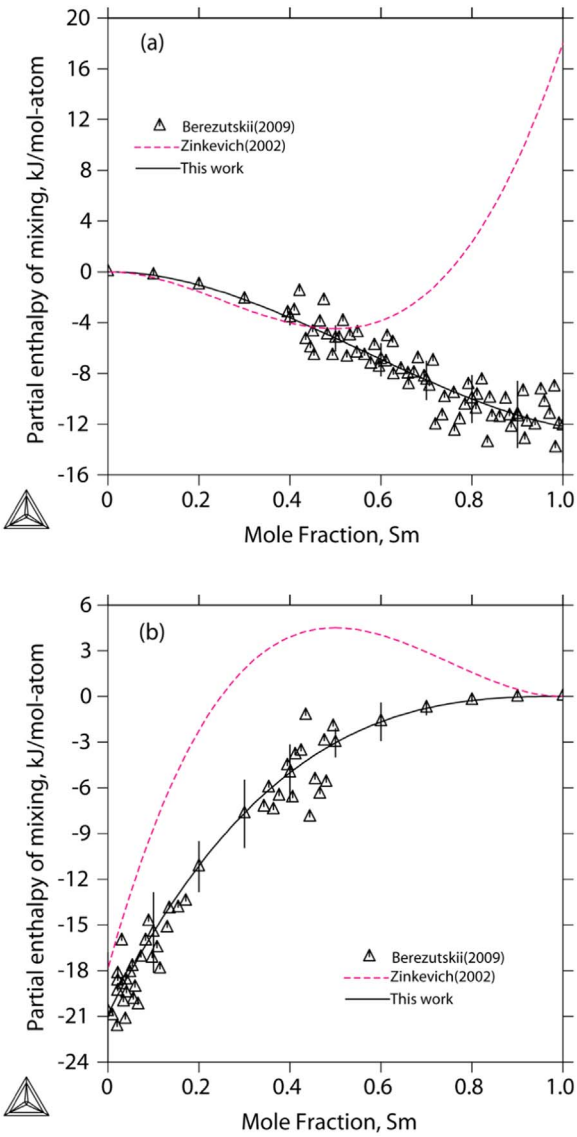

Fig. 10. Calculated partial enthalpy of Fe and Sm in liquid Fe-Sm alloys at 1829 K with the experimental data [26] and calculated results [24]. (Reference states: liquid Fe and liquid Sm).

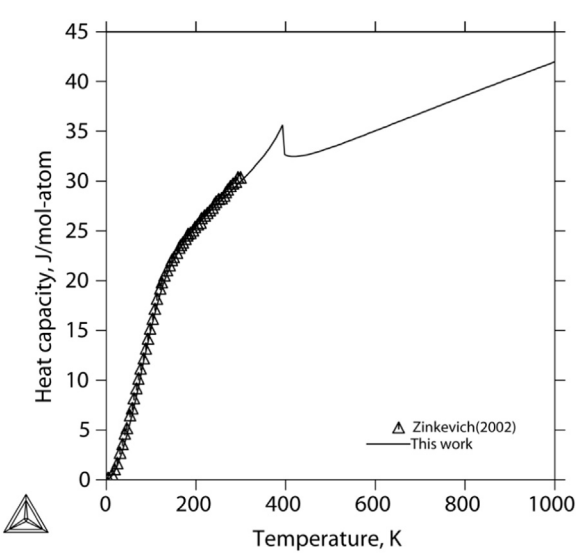

Fig. 11. Calculated heat capacity of $Fe_{17}Sm_2$ with the experimental data [24].

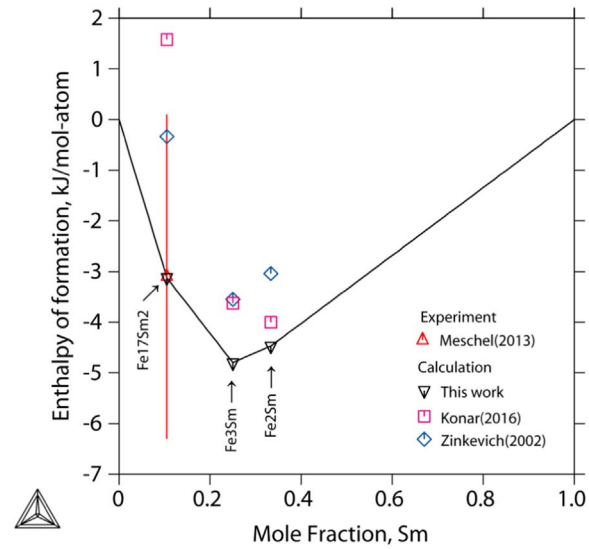

Fig. 12. Calculated enthalpies of formation of intermetallic compounds at 298 K with the experimental data [20] and the calculated results [24,27]. (Reference states: α-Fe and α-Sm).

The calculated results by Zinkevich et al. [24] show the intermetallic compound $Fe_{17}Sm_2$ is decomposed abnormally at about 300 K, while three intermetallic compounds, $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm, are stable at low temperature in the present calculation. Fig. 8 compares the calculated Fe-Sm binary phase diagram in this work with the experimental results [25]. The calculated results show good agreement with the experimental data [25]. Table 5 shows the calculated temperatures and compositions of invariant reactions in the Fe-Sm binary system with the experimental data. In this work, the calculated temperatures of the eutectic reaction (L ↔ Fe_2Sm + α-Sm) and three peritectic reactions (L + γ-Fe ↔ $Fe_{17}Sm_2$, L + $Fe_{17}Sm_2$ ↔ Fe_3Sm and L + Fe_3Sm ↔ Fe_2Sm) are 995 K, 1557 K, 1287 K and 1174 K, respectively, which is in good agreement with the experimental data [25] within experimental error.

Fig. 9 is the calculated enthalpies of mixing of liquid Fe-Sm alloys at 1829 K with the experimental data measured by Berezutskii et al. [26] and the calculated results [24]. The partial enthalpies of mixing of Fe and Sm in liquid alloys at 1829 K with the experimental data [26] and the calculated results [24] are shown in Fig. 10(a) and Fig. 10(b), respectively. The calculated results in this work agree well with the experimental data [26] within experimental error. As can be seen in Fig. 9 and Fig. 10, the enthalpy of mixing and the partial enthalpy of mixing in liquid Fe-Gd alloys at 1829 K calculated by Zinkevich et al. [24] are obvious discrepancy with the experimental data [26].

Fig. 11 is the calculated heat capacity of the intermetallic compound $Fe_{17}Sm_2$ in this work with the experimental data [24]. The calculated results are good consistent with the experimental data determined by Zinkevich et al. [24]. The calculated entropy of formation of $Fe_{17}Sm_2$ at 298 K is 28.90 J/mol-K-atom, which is slight difference with the data (36.6 ± 1.5 J/mol-K-atom) measured by Zinkevich et al. [24]. The calculated enthalpies of formation of $Fe_{17}Sm_2$, Fe_3Sm and Fe_2Sm at 298 K with the experimental data [20] and the calculated results [24,27] are shown in Fig. 12. The positive enthalpy of formation of $Fe_{17}Sm_2$ assessed by Konar et al. [27] is completely opposite with the experimental data measured by Meschel et al. [20]. The calculated enthalpy of formation of $Fe_{17}Sm_2$ by Zinkevich et al. [24] is much less negative than the experimental data [20]. In this work, the calculated enthalpy of formation of $Fe_{17}Sm_2$ is −3.13 kJ/mol-atom, which agrees well with the experimental data (−3.10 ± 3.20 kJ/mol-atom) [20]. The calculated enthalpies of formation of Fe_3Sm and Fe_2Sm in this work are −4.80 and −4.47 kJ/mol-atom, respectively, which are more negative than the calculated values [24,27]. It was also noted that the enthalpy

of formation Fe_3Sm is more negative than that of Fe_2Sm, which is different with the assessed results in the Fe-Gd binary system in the present work. The enthalpies of formation of intermetallic compounds in the Fe-RE binary systems are still necessary to be measured experimentally to verify the calculated results.

5. Conclusions

On the basis of the critical review of the previous optimizations and experimental information on phase diagram and thermodynamic properties in the published literature, thermodynamic re-assessment of the Fe-Gd and Fe-Sm binary systems was performed using the CALPHAD method. Self-consistent and reliable thermodynamic parameters were obtained finally, which can reproduce satisfactorily most of the experimental data including both phase diagram and thermodynamic properties. Thermodynamic parameters of all phases in the Fe-Gd and Fe-Sm binary systems were obtained finally. It is fundamental to develop the compatible and available thermodynamic database of the RE-Fe-B-based alloys, which is very useful to investigate the relations between alloy composition, microstructure and magnetic properties of novel RE-Fe-B-based permanent magnets.

Acknowledgements

This work was supported by the National Basic Foundation of China (2014CB643703), the National Key Research and Development Program of China (2016YFB0700901) and the Guangxi Natural Science Foundation (2016GXNSFDA380015, 2016GXNSFGA380001, 2014GXNSFBA118235). The authors also acknowledge the National Natural Science Foundation of China (51461013), the Research Foundation of Guangxi Education Department (KY2016YB166) and the Guangxi Key Laboratory of Information Materials, Guilin University of Electronic Technology, China (151009-Z) for financial support.

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